AI Fundamentals and Workflow

Simple definitions • Core concepts • Practical AI pipeline

1. AI Fundamentals

Artificial Intelligence (AI): The ability of computer systems to perform tasks that usually require human intelligence, such as understanding language, recognizing patterns, and solving problems.

Machine Learning (ML): A branch of AI in which models learn patterns from data to make predictions or decisions.

Deep Learning (DL): A branch of ML that uses multi-layer neural networks to learn complex patterns.

Natural Language Processing (NLP): Techniques that help computers process, understand, and generate human language.

Generative AI: AI that creates new content such as text, images, audio, video, or code.

Large Language Model (LLM): A model trained on large amounts of text to understand and generate language.

Prompt: The instruction or question given to an AI model.

Token: A small unit of text processed by a language model; it may be a word, part of a word, or punctuation.

Embedding: A numerical representation of content that helps systems compare semantic meaning.

Hallucination: When an AI model produces information that sounds plausible but is incorrect or unsupported.

2. AI Workflow

1. Data collection: Gather relevant data from files, databases, APIs, sensors, or other sources.

2. Data preparation: Clean, organize, transform, and split data for the task.

3. Model training: Use an algorithm to learn patterns from prepared data (for traditional ML).

4. Evaluation: Test model quality with suitable metrics and examples before release.

5. Deployment: Make the model available through an application, API, or service.

6. Monitoring: Track performance, errors, safety, and changing data; update when needed.

3. RAG (Retrieval-Augmented Generation)

RAG combines an LLM with relevant information retrieved from external documents or a knowledge base. It can help provide more grounded, context-specific answers.

Typical RAG pipeline: Load documents → split into chunks → create embeddings → store in a vector database → retrieve relevant chunks for a query → add them to the prompt → generate an answer.

Vector database: Stores embeddings and helps retrieve semantically similar content. Examples include FAISS and Pinecone.

4. Where AI Is Used

Chatbots and virtual assistants • Recommendations • Image and speech recognition • Healthcare support • Fraud detection • Document search • Content generation.

Remember: traditional ML often learns from training examples; RAG retrieves relevant knowledge at question time to help an LLM answer with context.

AI Fundamentals and Workflow
A Simple Guide

1 What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) refers to the ability of machines or computer systems to perform tasks that typically require human intelligence, such as understanding language, recognizing patterns, learning from data, making decisions, and solving problems.

2 Key Concepts in AI

- **Data:** The information used to train or run AI models.
- **Algorithms:** Set of rules or methods used to find patterns and make predictions.
- **Model:** A program that learns from data and makes predictions or decisions.
- **Training:** The process of teaching a model using data.
- **Inference:** Using a trained model to make predictions on new data.
- **Evaluation:** Measuring how well the model performs.

3 Types of AI

- **Narrow AI (Weak AI):** Designed for specific tasks (e.g., chatbots, recommendation systems).
- **General AI (Strong AI):** Hypothetical AI with human-like intelligence (still under research).
- **Generative AI:** Creates new content like text, images, code, etc. (e.g., ChatGPT, DALL·E).

4 Important AI Terms

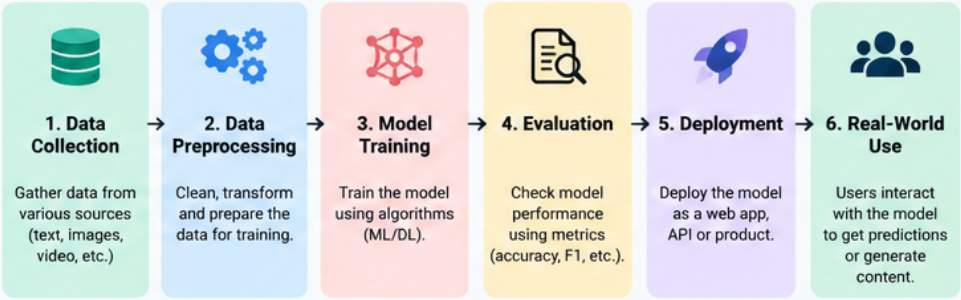
- **Machine Learning (ML):** Subset of AI that learns from data without explicit programming.
- **Deep Learning (DL):** Subset of ML using neural networks to learn complex patterns.
- **Natural Language Processing (NLP):** Helps machines understand and process human language.
- **Computer Vision:** Helps machines understand images and videos.
- **Large Language Model (LLM):** A type of deep learning model trained on large text data.
- **Generative AI:** AI that generates new content (text, image, audio, code, etc.).
- **RAG (Retrieval Augmented Generation):** Combines external knowledge with LLM for accurate and up-to-date responses.

5 Real-World Applications

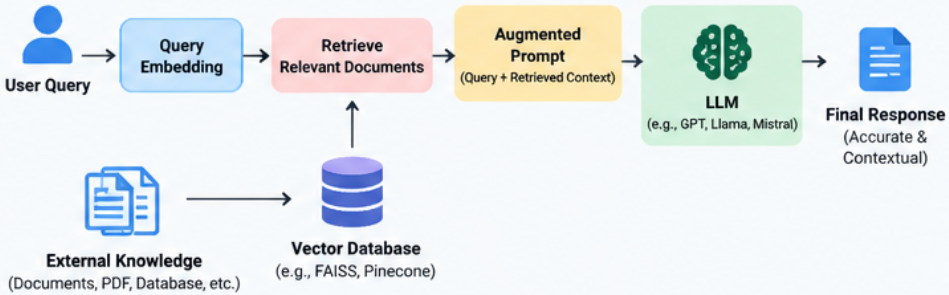
- Chatbots and Virtual Assistants (Customer Support)
- Recommendation Systems (E-commerce, OTT platforms)
- Image and Speech Recognition (Face unlock, Voice assistants)
- Healthcare (Disease prediction, Medical image analysis)
- Finance (Fraud detection, Risk analysis)
- Education (Personalized learning)
- Content Creation (Text, Image, Video generation)

AI Workflow with Architecture
From Data to Real-World Application

1 AI Workflow (End-to-End Process)



2 RAG (Retrieval Augmented Generation) Architecture



3 Example: Real-World Application (Chatbot with RAG)

